

This protocol outlines the procedure for DNA digestion using USER enzyme and subsequent Qubit quantification.

1. Prepare Digestion Mix: Add 1 uL of USER enzyme to 49 uL of TE buffer.
2. Pool Beads: Add 5 µL of 1X B&W buffer to exactly 20 synthesis tubes to resuspend the beads. Transfer the suspension from all 20 tubes into a single collection tube.
3. Wash: Place the tube on a magnetic rack, remove the supernatant, and rinse the beads once with TE Buffer.
4. Add First Mix: Remove the supernatant and take the tube off the magnetic rack. Add 20 uL of the Digestion Mix.
5. Incubate: Mix well and place on a heatershaker at 37°C and 1000 rpm.
6. First Reaction: Incubate for 30 minutes. Add 0.4 uL of USER enzyme, mix, and incubate for another 30 minutes. Mix again, then incubate for a final 30 minutes.
7. First Collection: Place on the magnetic rack. Transfer the supernatant to a collection tube.
8. Add Second Mix: Remove the tube from the magnet. Add 15 uL of the Digestion Mix to the beads.
9. Second Reaction: Repeat the incubation process (Step 6).
10. Final Collection: Place on the magnetic rack. Collect the supernatant and combine it with the first collection to get the final product solution.
11. Prepare Qubit Working Solution: Add 1 uL of Qubit Reagent to 199 uL of Qubit Buffer. Mix well by pipetting to prepare the working solution.
12. Sample Dilution: In a 200 µL tube, add 1 µL of the final product solution to 199 µL of the Qubit Working Solution. Mix well by pipetting.
13. Cap and Centrifuge: Seal the tube tightly and centrifuge briefly to ensure the liquid is at the bottom and bubbles are removed.
14. Incubate in Dark: Incubate the mixture in the dark at room temperature for 2 minutes.
15. Measurement: Place the tube in the fluorometer and measure the concentration.